FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°6_ – RÉSEAUX 2

COURS: RÉSEAUX 2

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NOM: PÉTIOTE

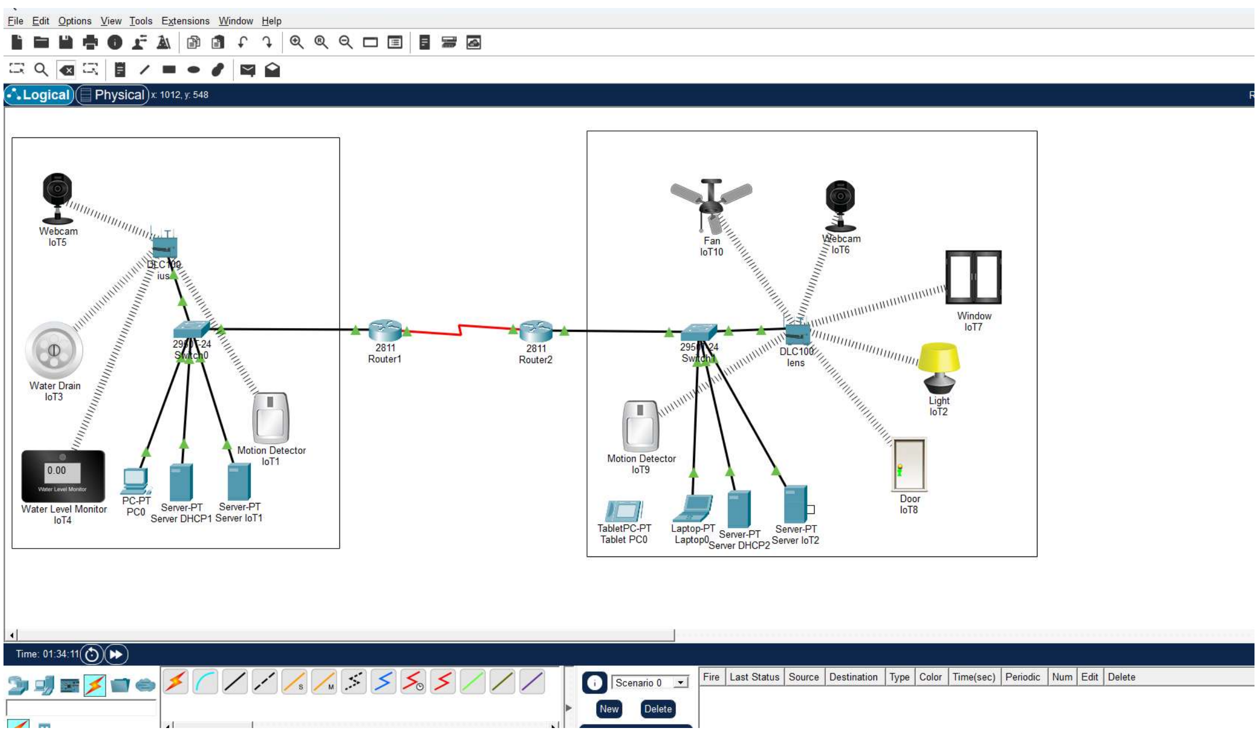
NIVEAU: L3

DATE: 04/05/2026

L’objectif de ce TD est de:

1. Le contrôle à distance
2. La communication entre objets
3. La sécurité du réseau IoT

Conception et configuration d’un réseau Smart Home (IoT)



Configuration des routeurs

R1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/2/0
Router(config-if)#ip address 192.168.1.1 255.255.255.252
Router(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config)#
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up
```

Ctrl+F6 to exit CLI focus

R2

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)## Configuration de l'interface srie
                 ^
% Invalid input detected at '^' marker.

Router(config)#interface Serial0/2/0
Router(config-if)#ip address 192.168.1.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)## Configuration de l'interface FastEthernet0/0 pour Smarthome
                 ^
% Invalid input detected at '^' marker.

Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up
```

Ctrl+F6 to exit CLI focus

Routage entre les deux routeurs (OSPF)

R1

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#do wr
00:08:45: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on
Serial0/2/0 from LOADING to FULL, Loading Done

Building configuration...
[OK]
Router(config-router)#
```

Ctrl+F6 to exit CLI focus   Copy   Paste

Top

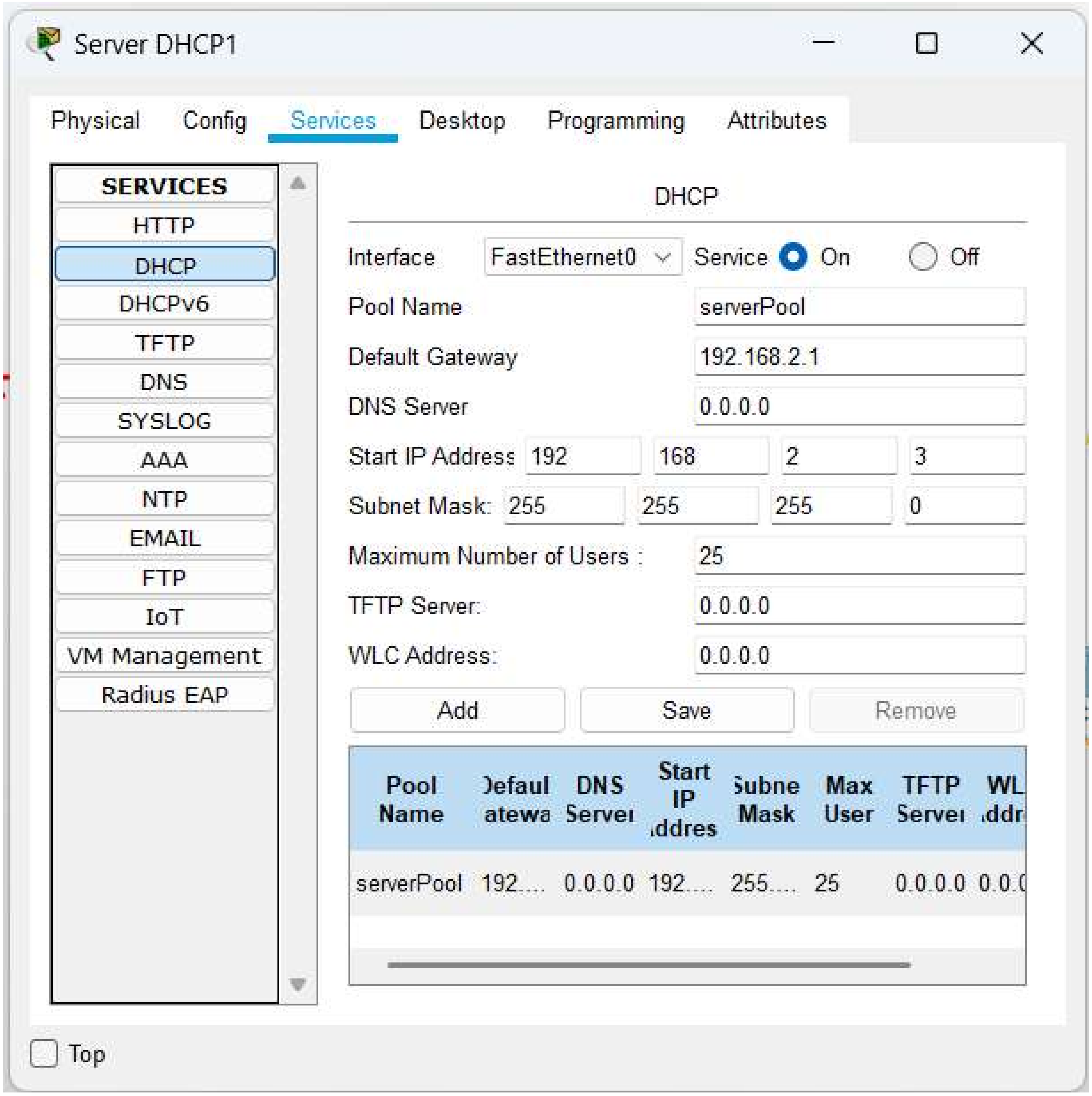
R2

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

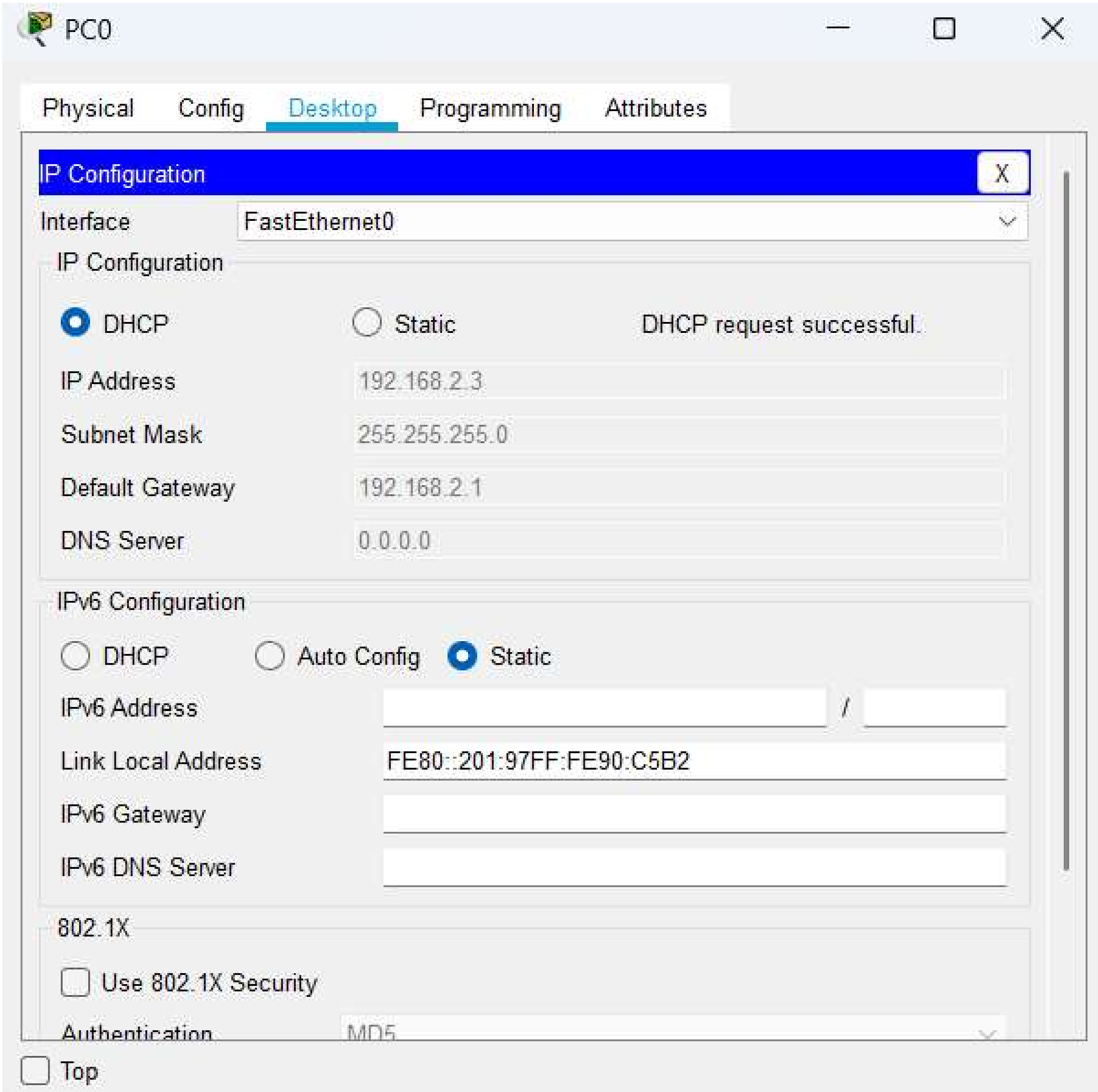
Ctrl+F6 to exit CLI focus   Copy   Paste

Top

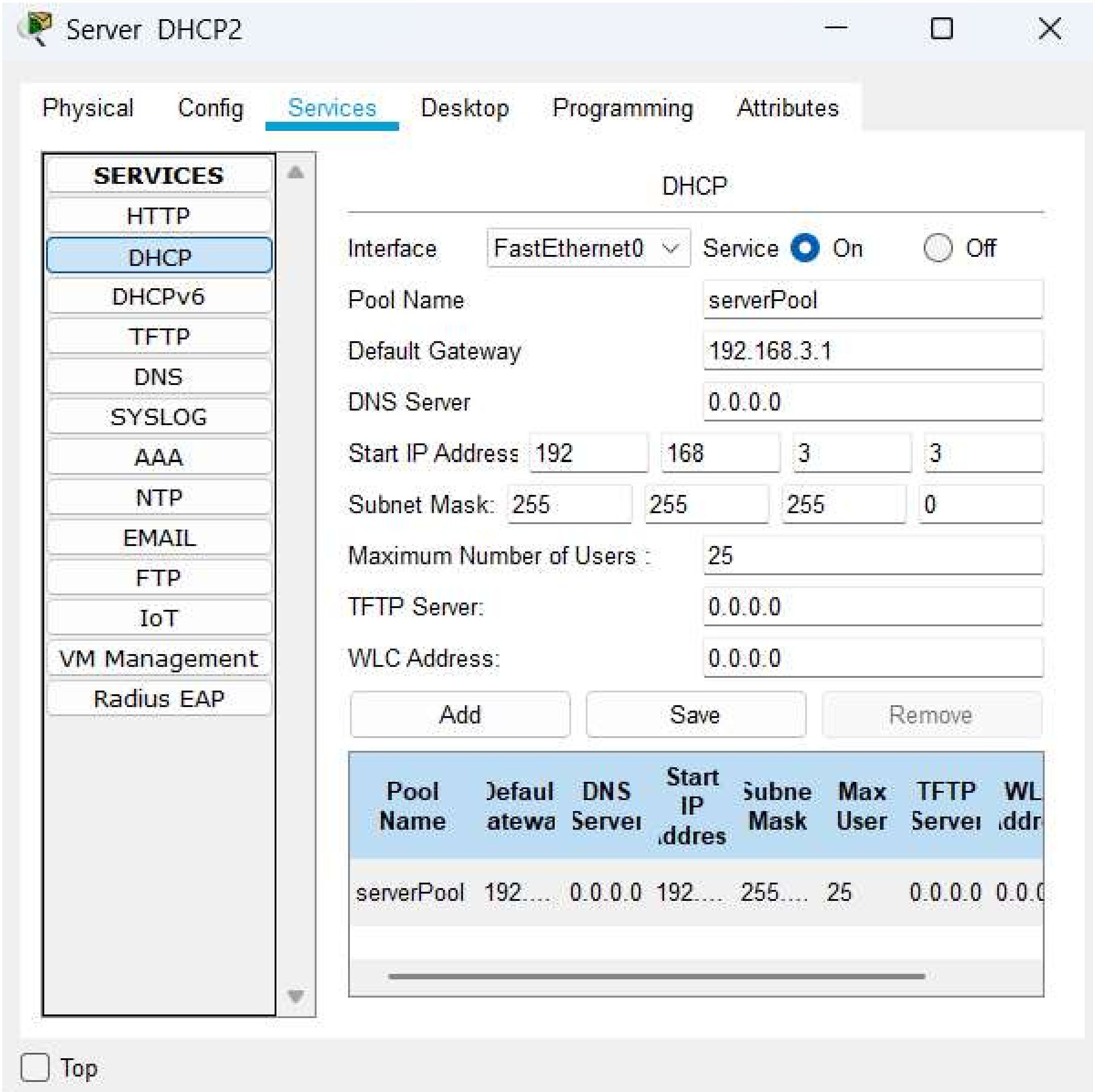
Configuration des servers DHCP pour attribuer des IP automatiquement


Server DHCP1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
192.168.2.1
DNS Server
0.0.0.0
Start IP Address
192
168
2
3
Subnet Mask:
255
255
255
0
Maximum Number of Users :
25
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Start IP
DNS Server
Max User
TFTP Server
serverPool
192....
0.0.0.0
192....
255....
25
0.0.0.0
Top

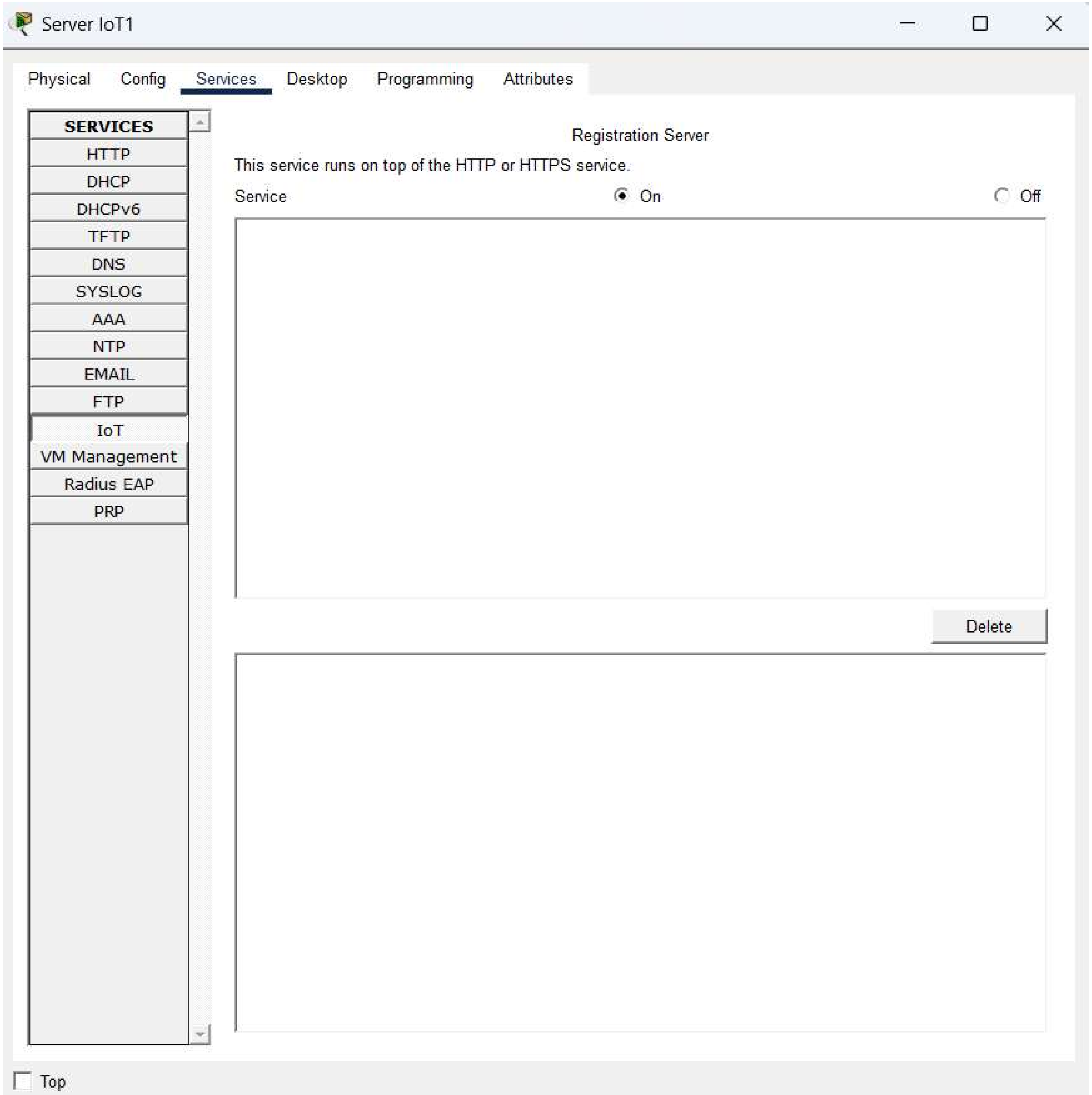
Exemple

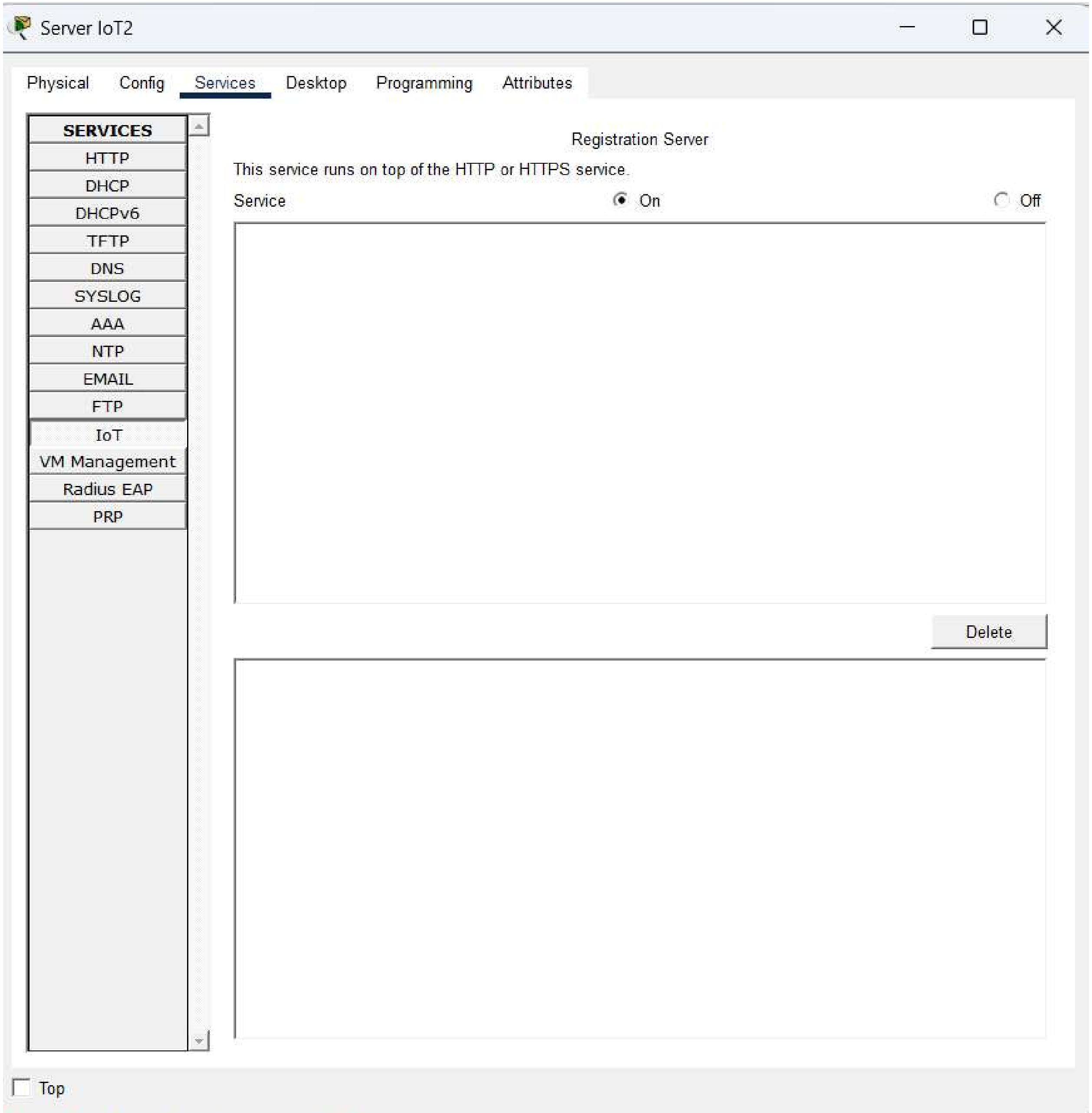

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IP Address
192.168.2.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address
FE80::201:97FF:FE90:C5B2
IPv6 Gateway
IPv6 DNS Server
802.1X
Use 802.1X Security
Top

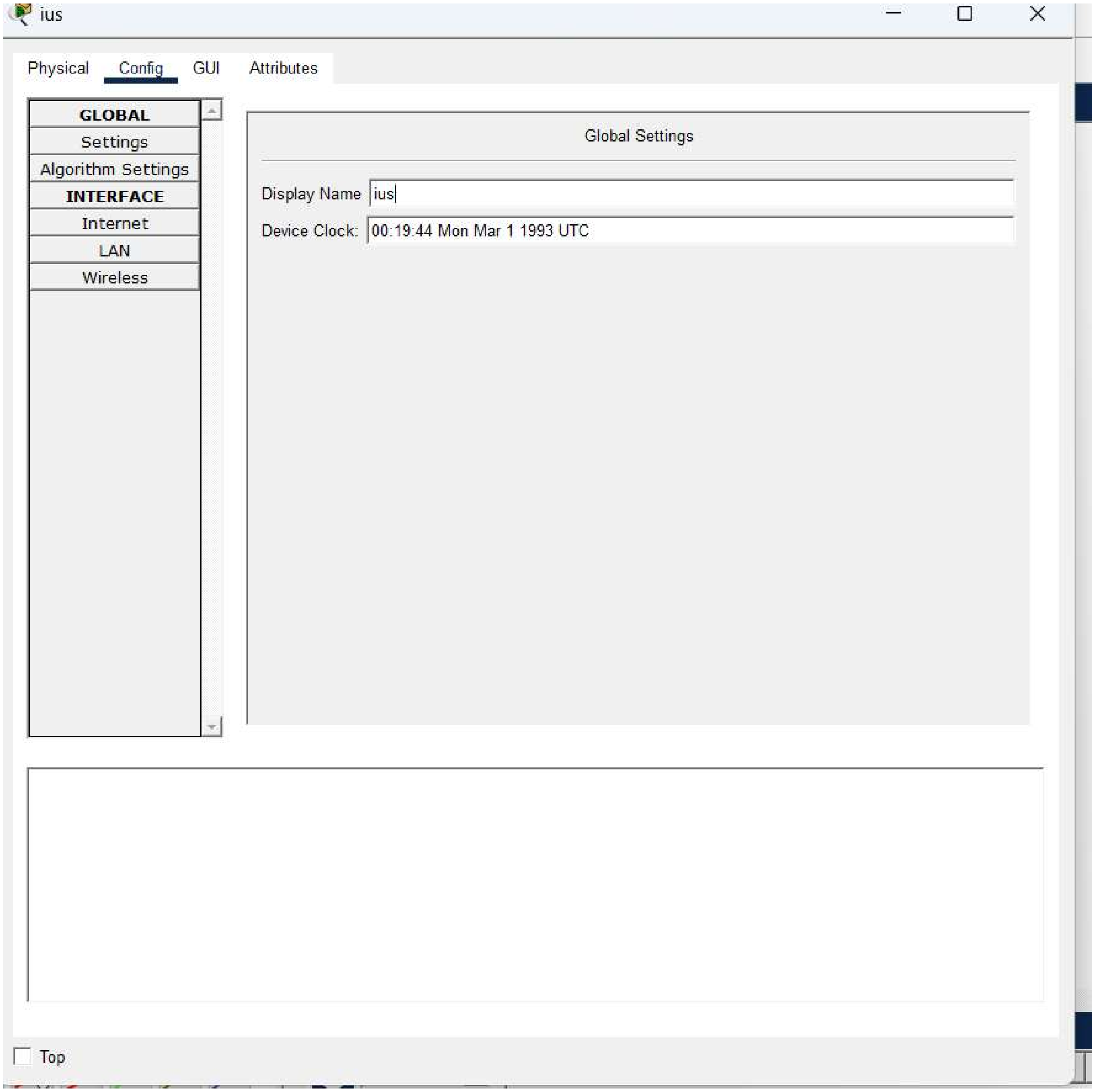
DHCP 2

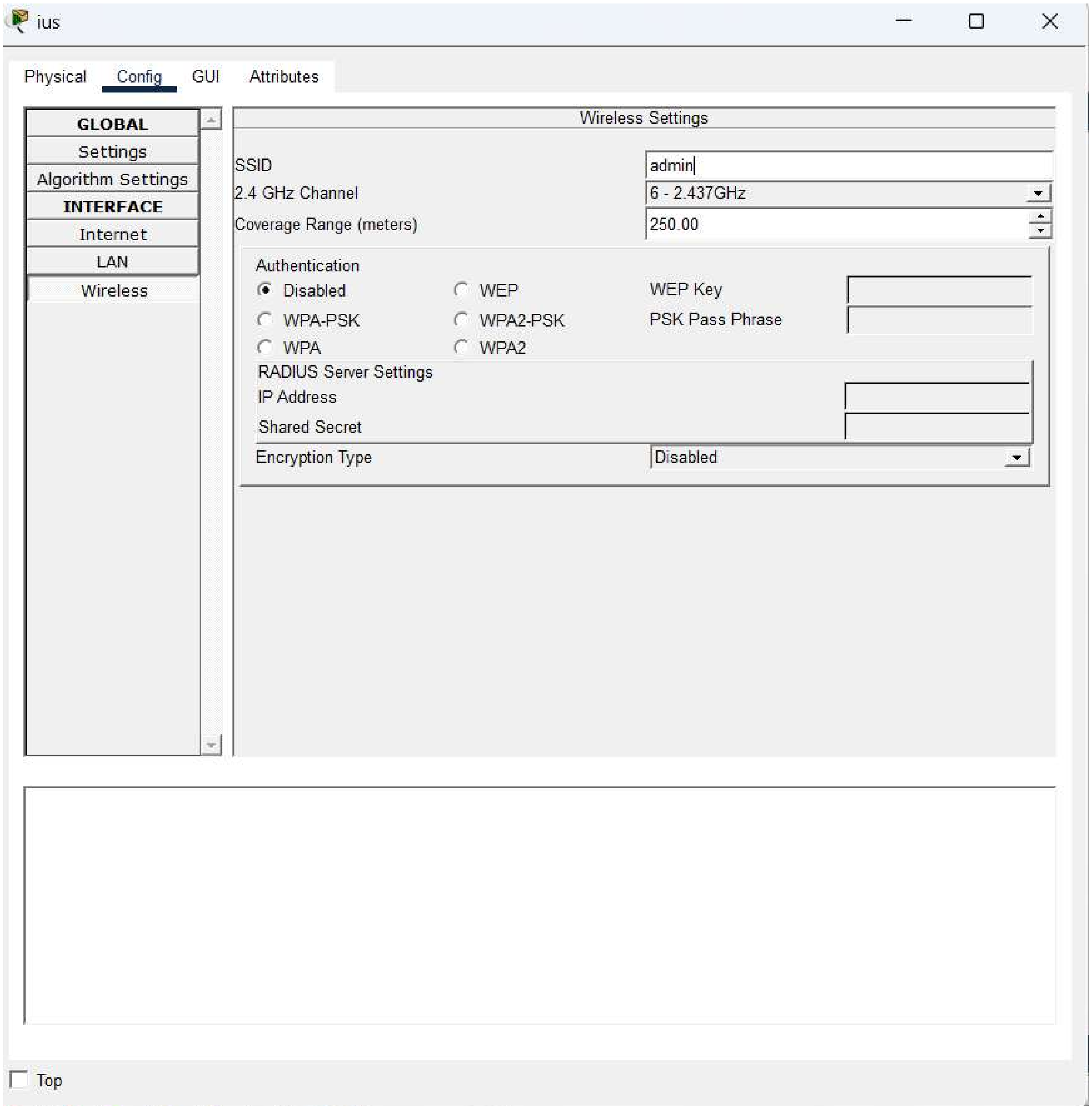


Configuration des servers Io

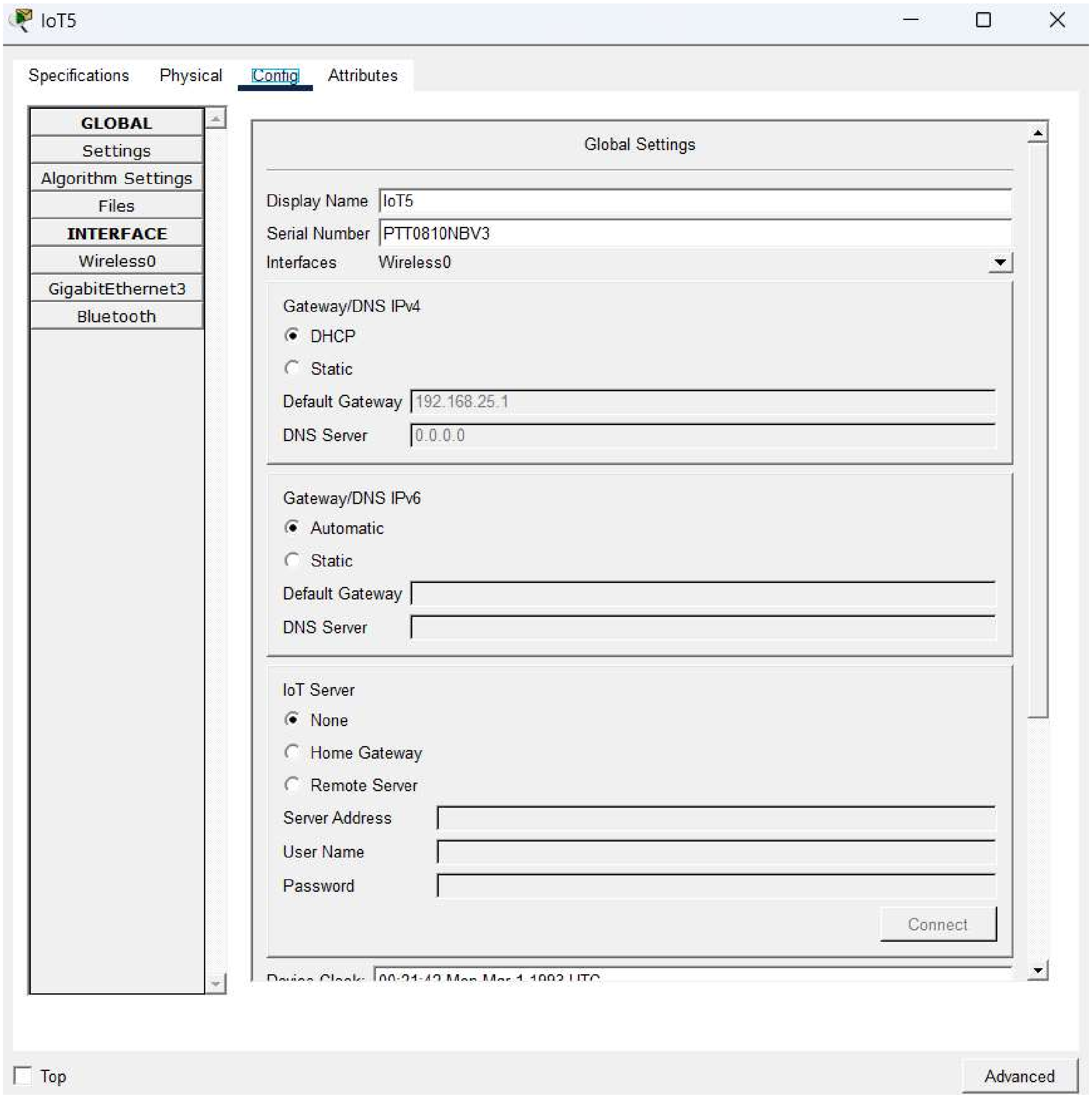

Server IoT1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
Registration Server
This service runs on top of the HTTP or HTTPS service.
Service
On
Off
Delete
Top


Server IoT2
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
Registration Server
This service runs on top of the HTTP or HTTPS service.
Service
On
Off
Delete
Top


ius
Physical
Config
GUI
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Internet
LAN
Wireless
Global Settings
Display Name
ius
Device Clock:
00:19:44 Mon Mar 1 1993 UTC
Top

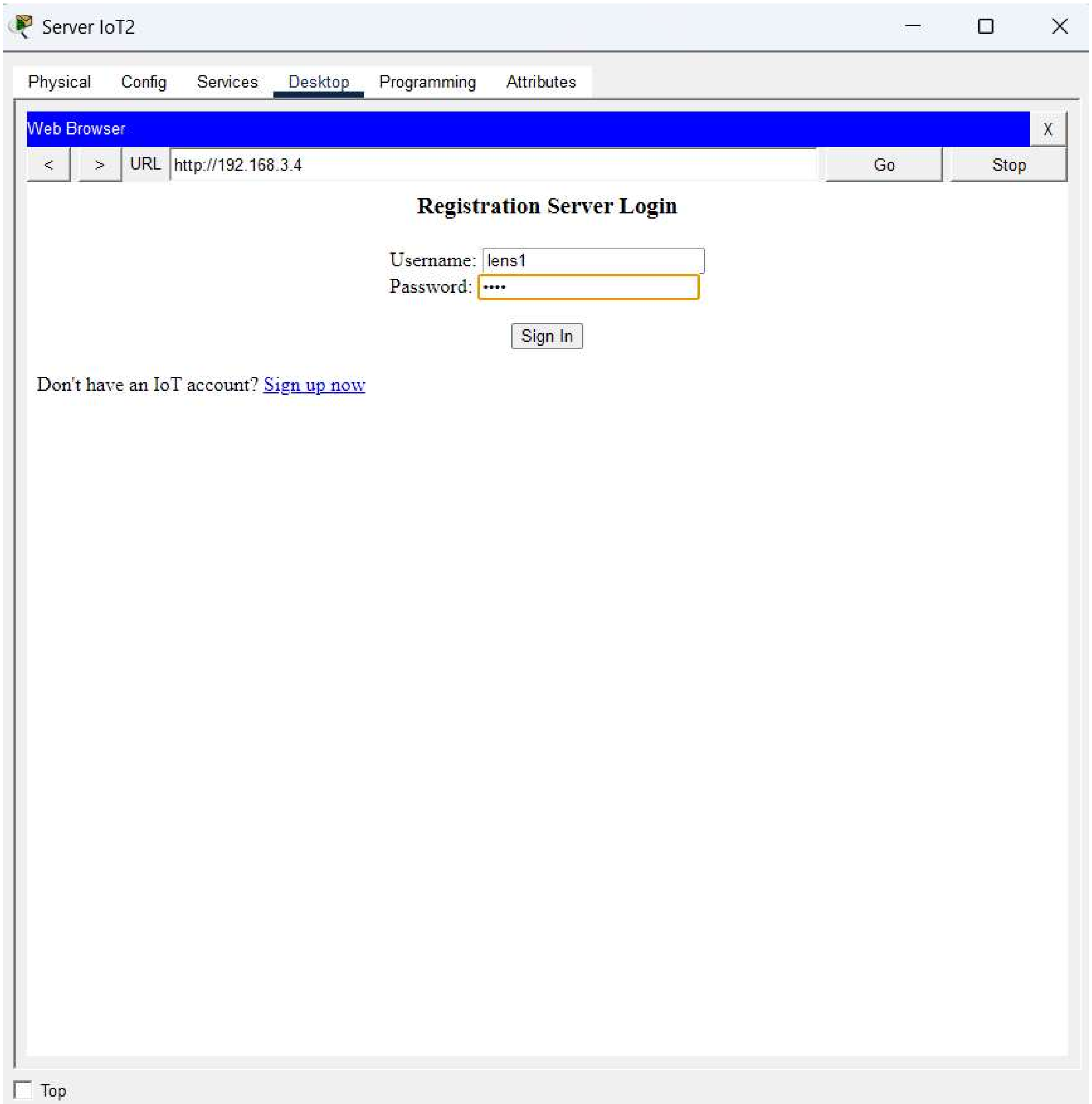


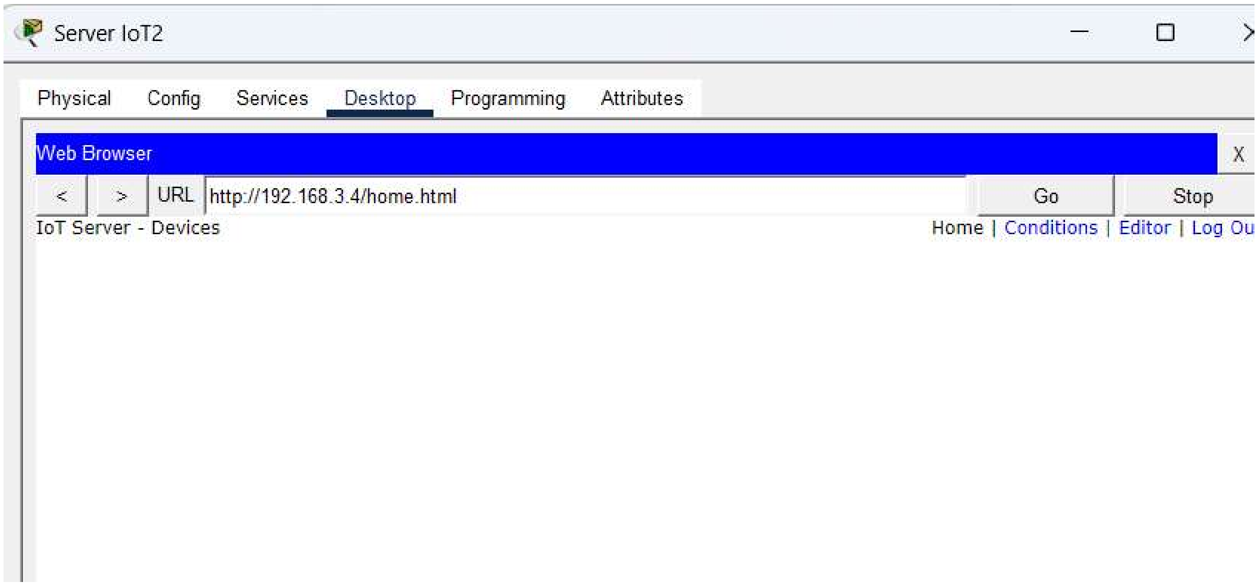
Exemple d’un appareil



Registration au server

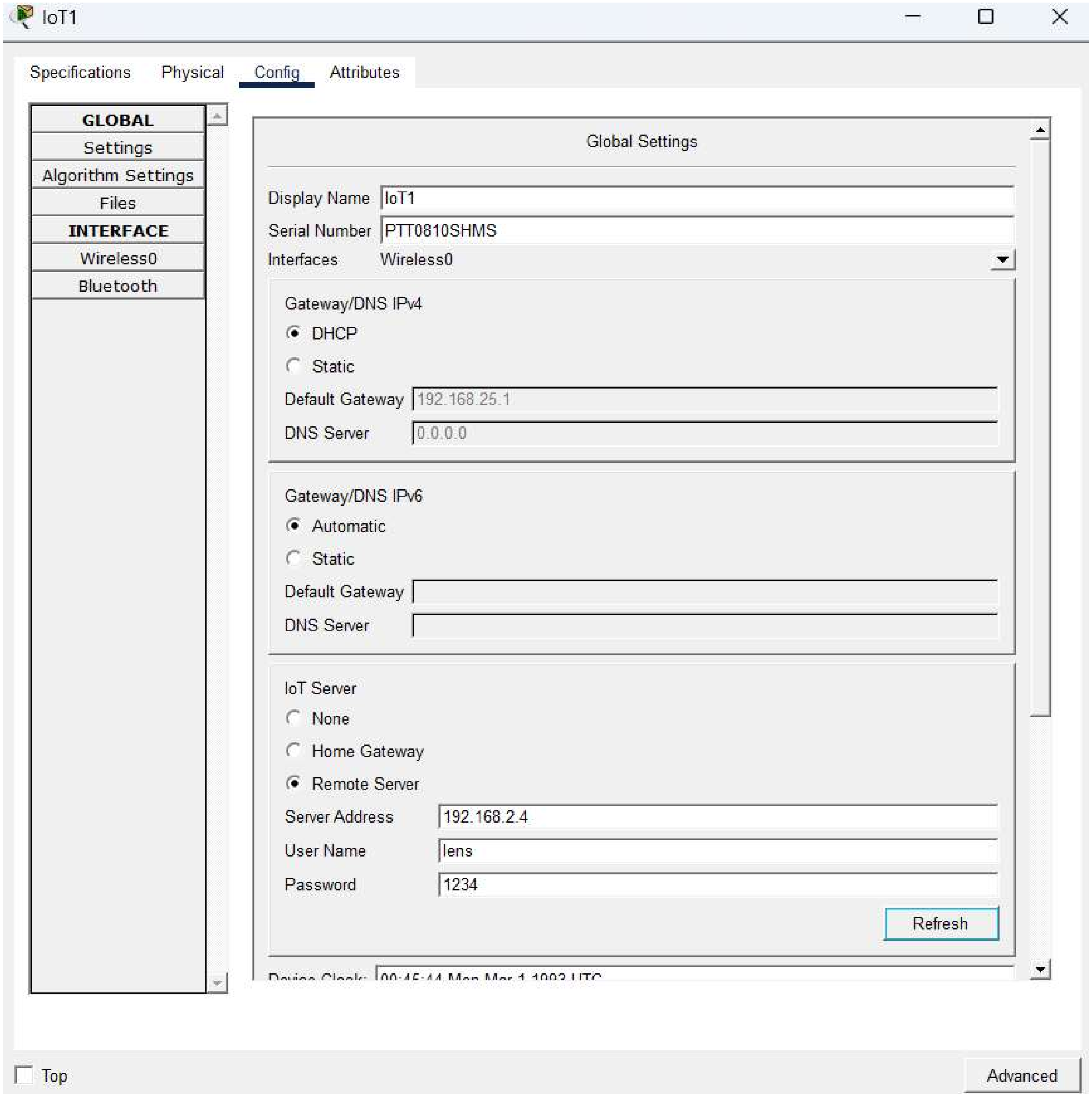
IoT 1

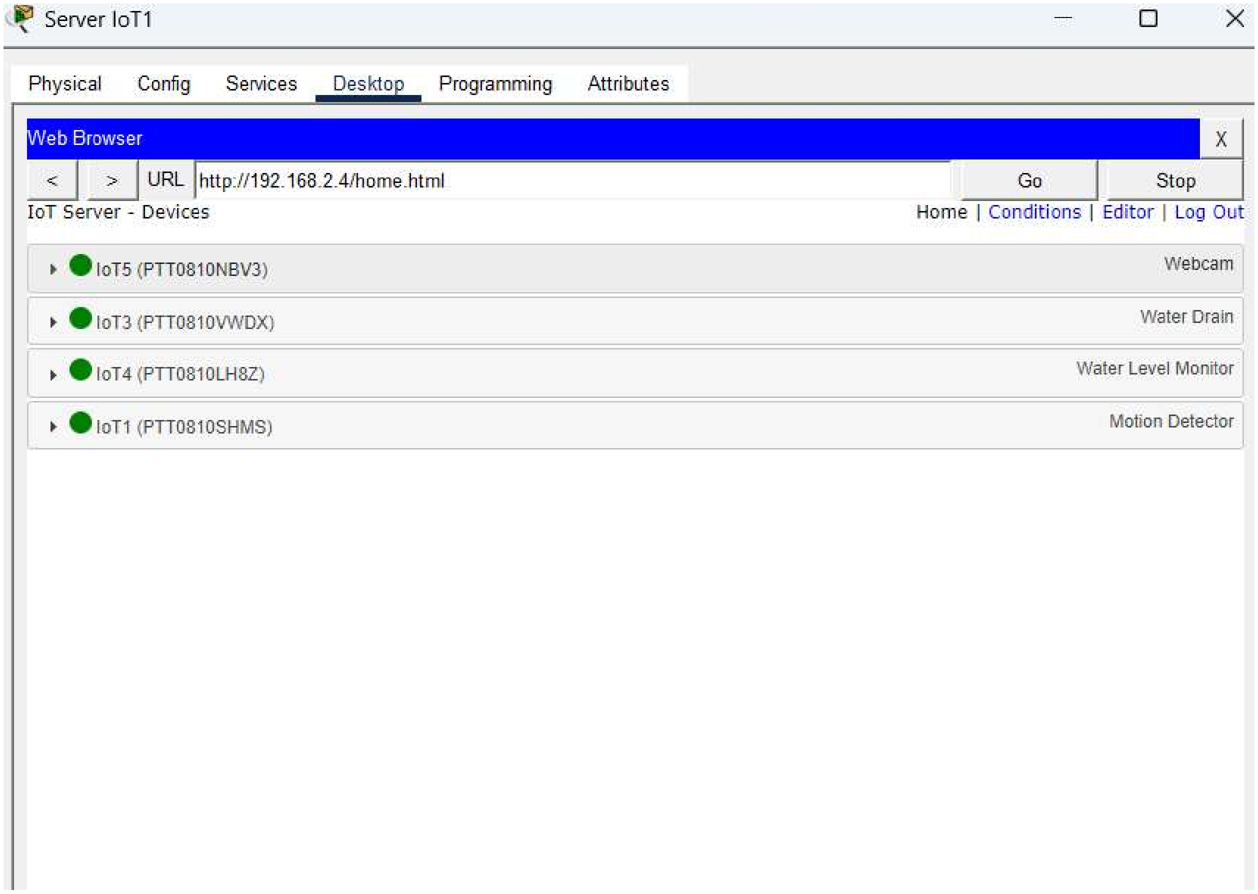

Server IoT2
Physical
Config
Services
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://192.168.3.4
Go
Stop
Registration Server Login
Username: lens1
Password: ••••
Sign In
Don't have an IoT account? Sign up now
Top



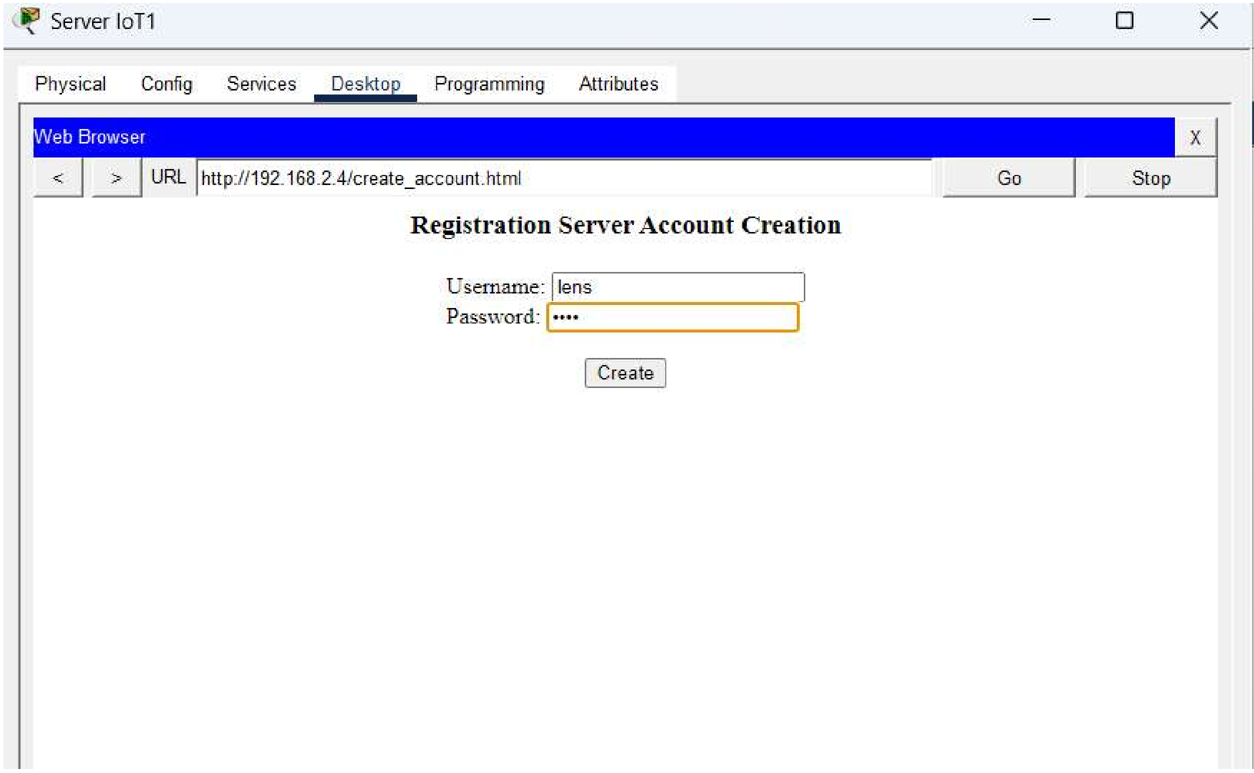
Ajout des IoT sur le serveur

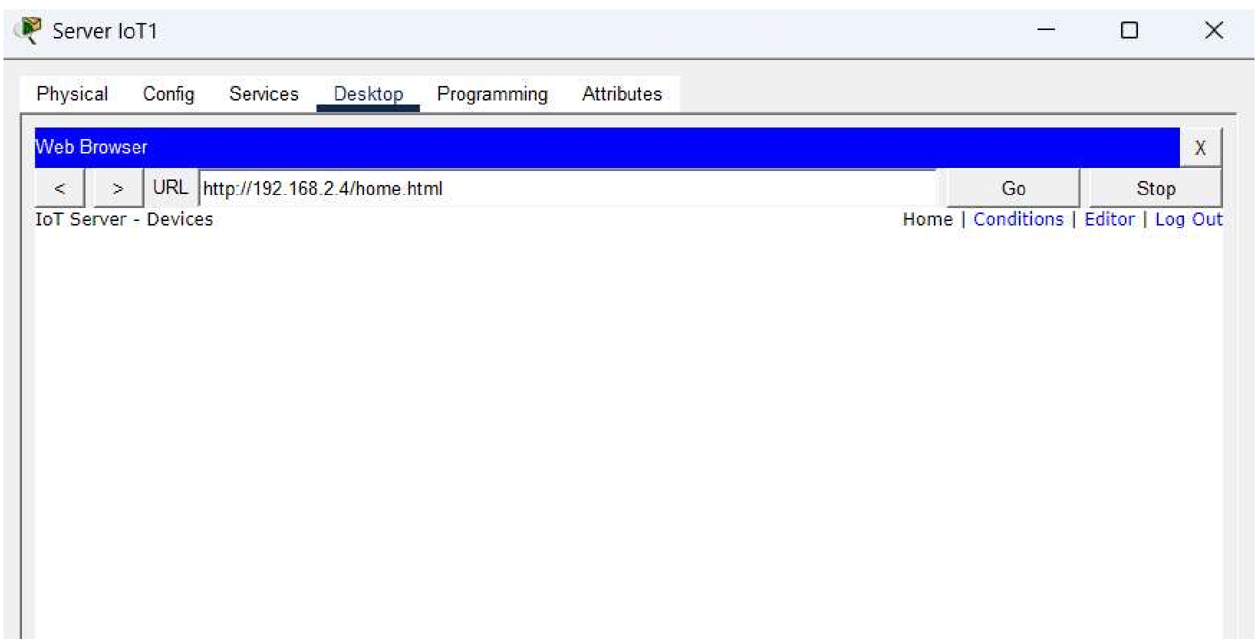
Exemple de configuration d'un appareil


IoT1
Specifications
Physical
Config
Attributes
GLOBAL
Settings
Algorithm Settings
Files
INTERFACE
Wireless0
Bluetooth
Global Settings
Display Name
IoT1
Serial Number
PTT0810SHMS
Interfaces
Wireless0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.25.1
DNS Server
0.0.0.0
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
IoT Server
None
Home Gateway
Remote Server
Server Address
192.168.2.4
User Name
lens
Password
1234
Refresh
Top
Advanced

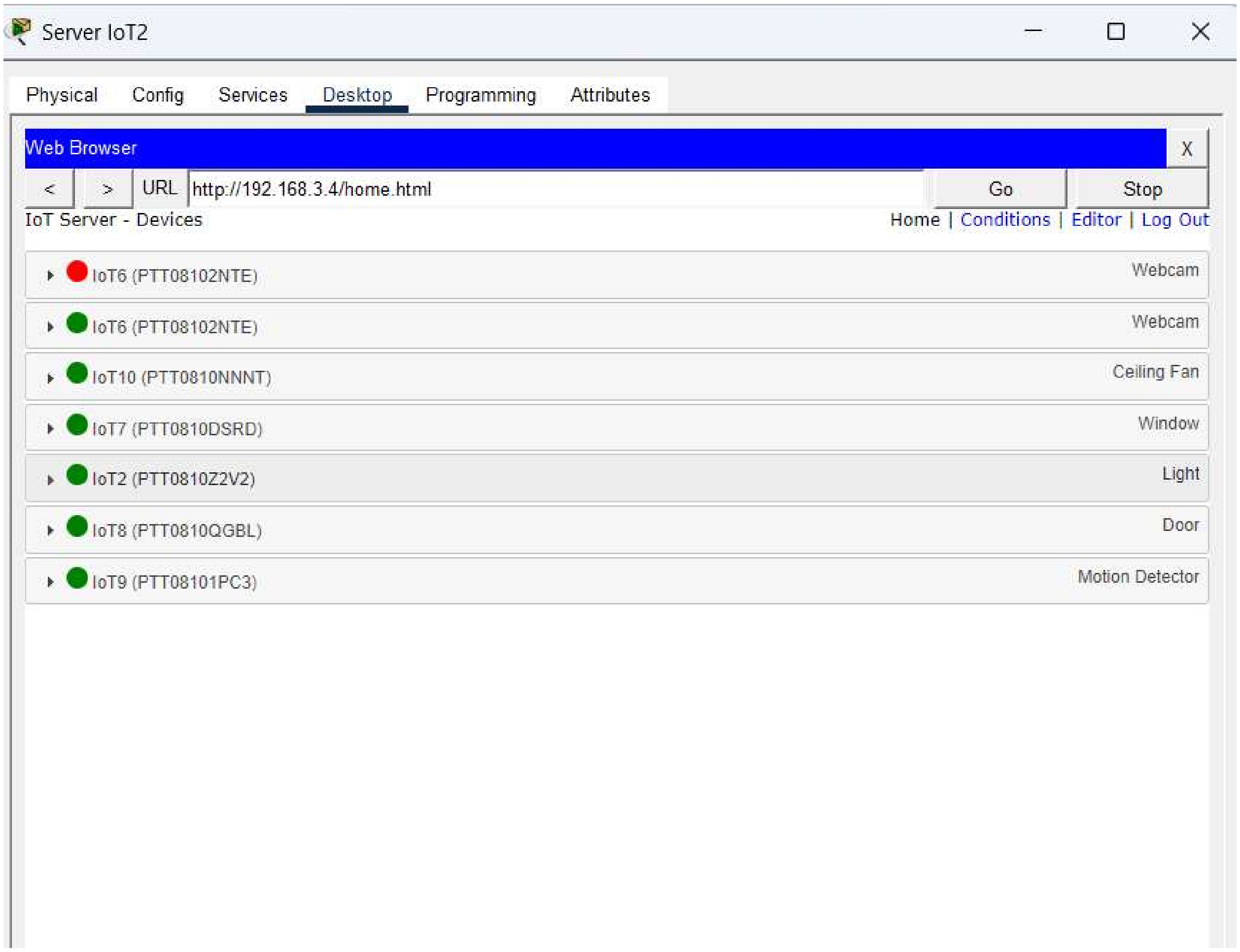

Server IoT1
Physical
Config
Services
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://192.168.2.4/home.html
Go
Stop
IoT Server - Devices
Home | Conditions | Editor | Log Out
IoT5 (PTT0810NBV3)
Webcam
IoT3 (PTT0810VWDX)
Water Drain
IoT4 (PTT0810LH8Z)
Water Level Monitor
IoT1 (PTT0810SHMS)
Motion Detector

IoT 2



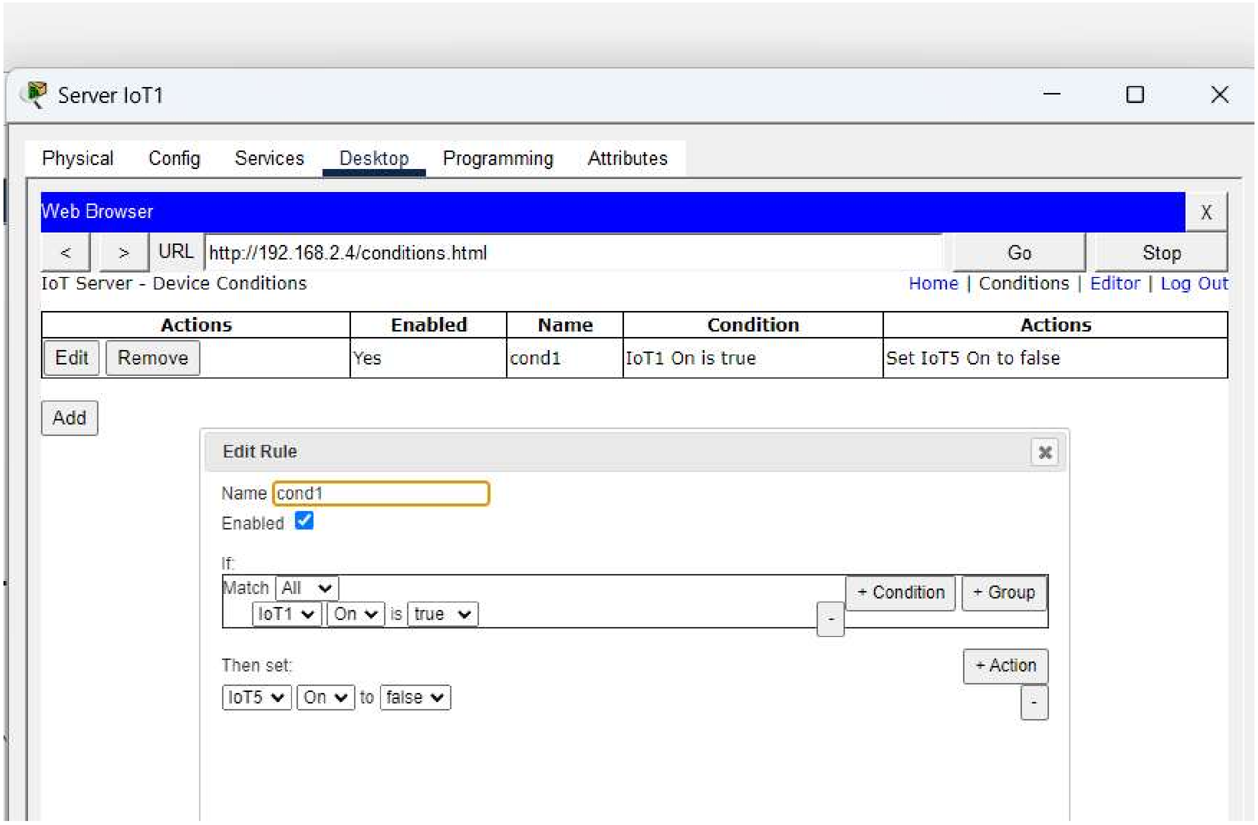


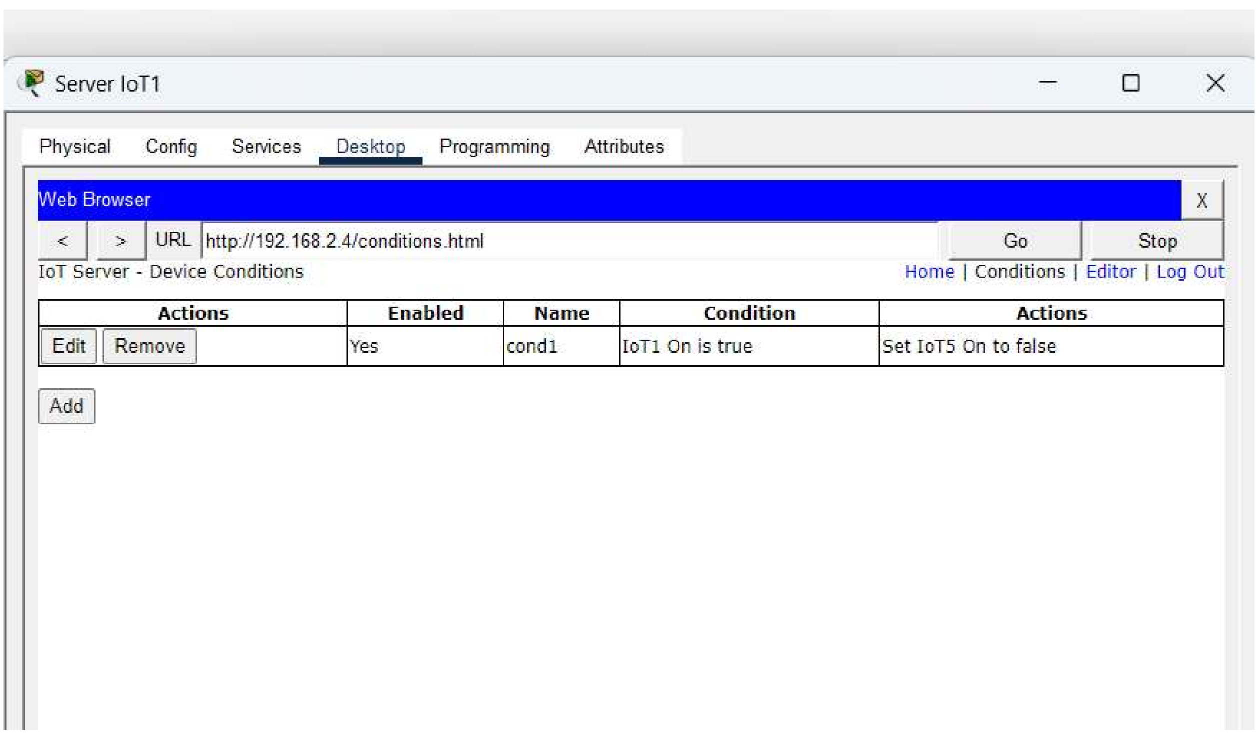
Ajout des IoT sur le serveur



Automatisation (IoT Rules)

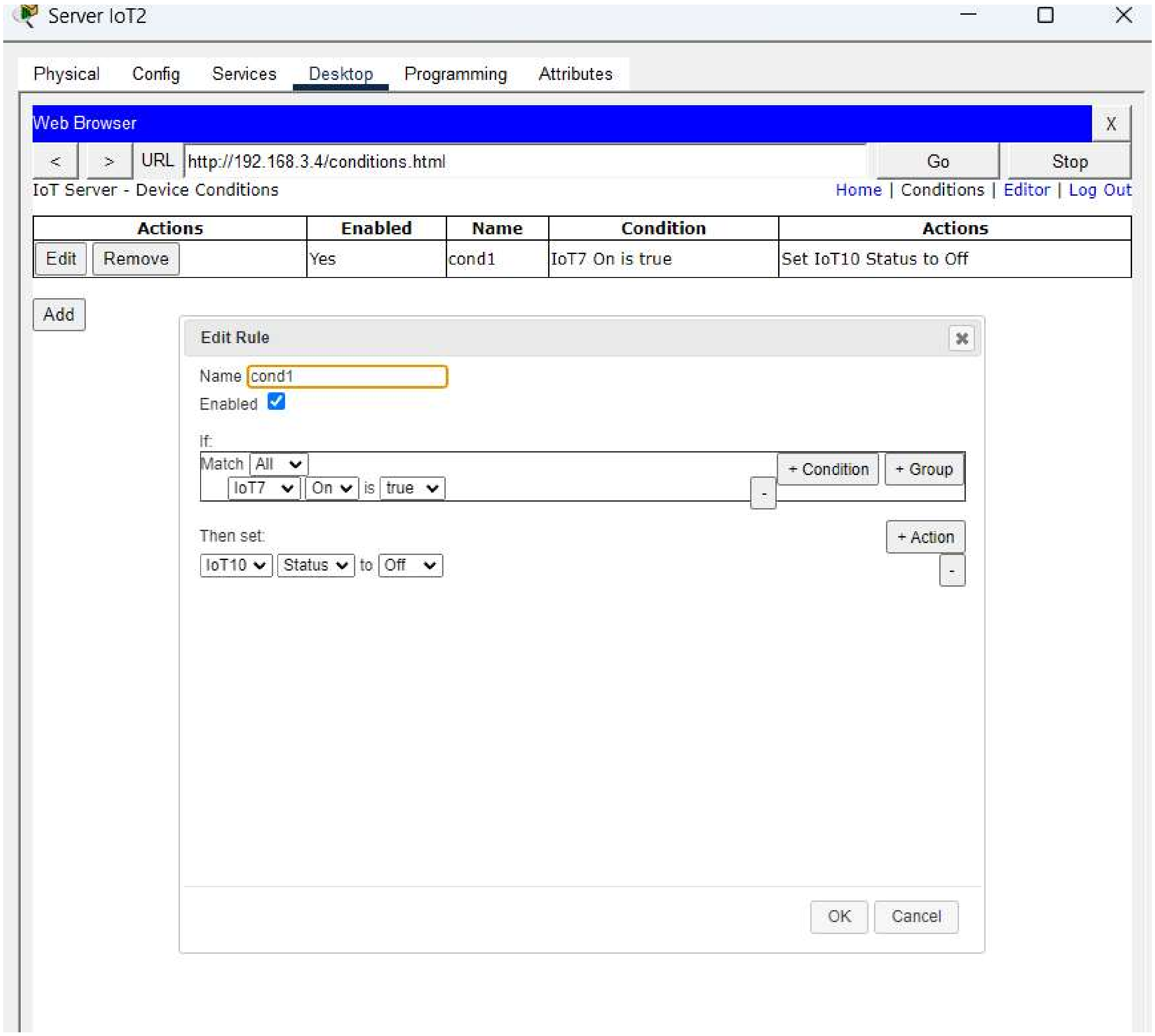
IoT 1





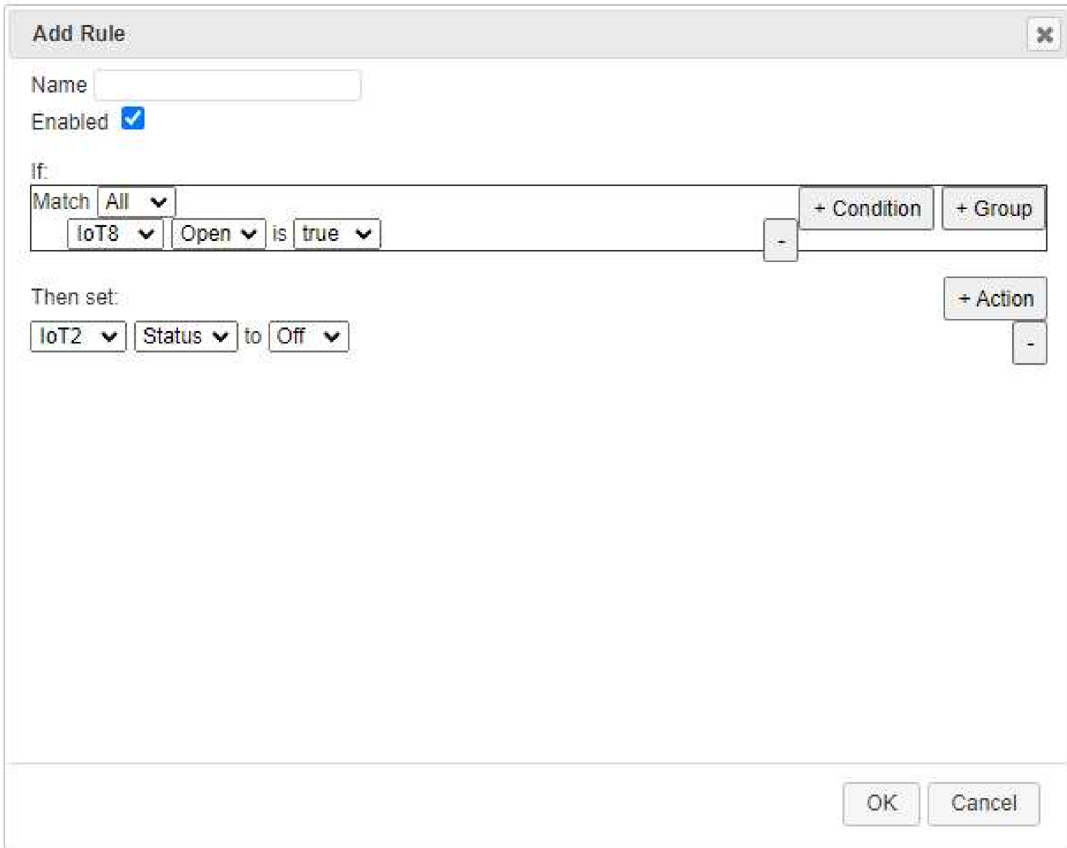
Notre condition: Si le motion detector est ouvert, le camera s'etteint.

IoT 2

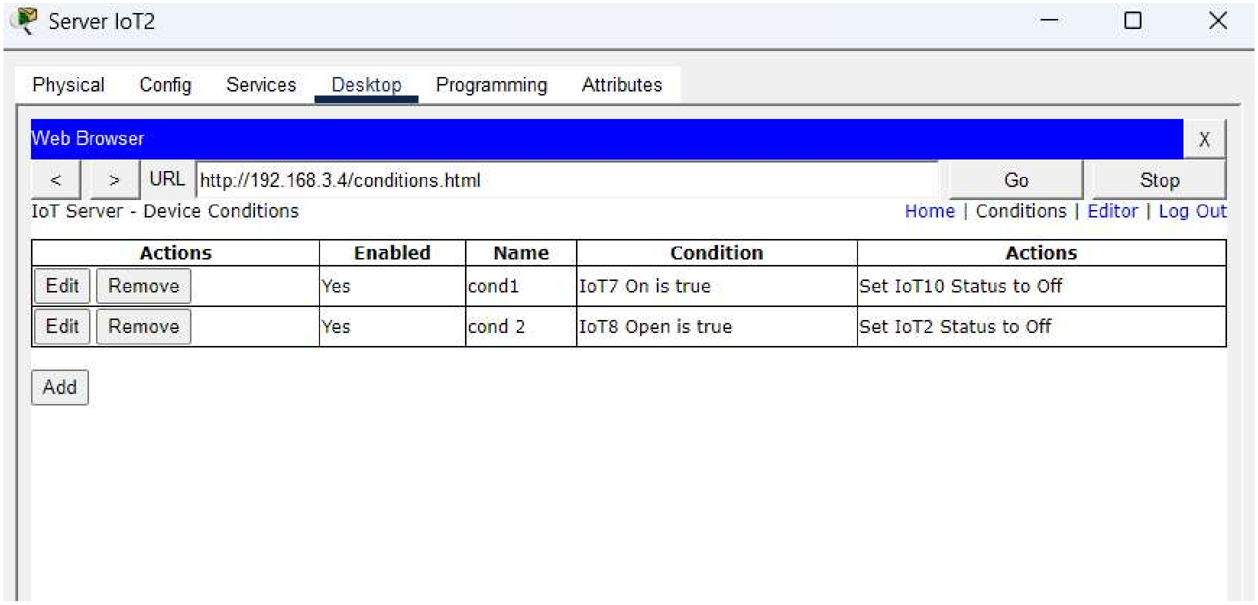


Si la fenetre est ouvert, le ventilateur doit s'etteindre.

Condition 2



Si la porte est ouvert la lampe doit s’etteindre



Coclusion.

Ce TD m’a permis de savoir comment faire la configuration des IOT, leurs sécurisations et la communication entre eux.donc la tache a été réussie.